Kaiyuan Ji

Ph.D. student
School of ECE, Cornell University
136 Hoy Road, Ithaca, NY 14850, USA

kj264@cornell.edu
kaiyuanji.github.io
Google Scholar

Education

Cornell University — Ithaca, New York, USA
Ph.D. in Electrical and Computer Engineering, advised by Mark M. Wilde — 08/2022 – present

Tsinghua University — Beijing, China
B.Eng. in Computer Science (Yao Class), advised by Xiongfeng Ma — 08/2018 – 06/2022

Research Interests

Quantum information theory: quantum Shannon theory, quantum hypothesis testing, entropic measures, quantum resource theories, quantum thermodynamics.

Foundations of quantum theory: causal structures of quantum processes, nonclassical correlations, interpretation of quantum theory, generalised and operational probabilistic theories.

Publications

Preprint Articles

[2] Kaiyuan Ji, Gilad Gour, and Mark M. Wilde, "Fundamental limits for thermodynamic control with quantum feedback", March 2026. [arXiv:2503.09012]

[1] Kaiyuan Ji and Bartosz Regula, "Beyond Hoeffding and Chernoff: Trading conclusiveness for advantages in quantum hypothesis testing", December 2025. [arXiv:2510.07601]

Journal Articles

[5] Kaiyuan Ji, Seth Lloyd, and Mark M. Wilde, "Retrocausal capacity of a quantum channel: Communicating through noisy closed timelike curves", *Physical Review Letters* **136**, 230801, June 2026. [arXiv:2509.08965]

[4] Kaiyuan Ji, Hemant K. Mishra, Milán Mosonyi, and Mark M. Wilde, "Barycentric bounds on the error exponents of quantum hypothesis exclusion", *IEEE Transactions on Information Theory* **72**, 2391, April 2026. [arXiv:2407.13728]

[3] Kaiyuan Ji and Eric Chitambar, "Entropic and operational characterizations of dynamic quantum resources", *IEEE Transactions on Information Theory* **71**, 7753, October 2025. [arXiv:2112.06906]

[2] Kaiyuan Ji, Bartosz Regula, and Mark M. Wilde, "Postselected communication over quantum channels", *International Journal of Quantum Information* **22**, 2440012, August 2024. Invited contribution to *Special Issue: Alexander S. Holevo's 80th birthday*. [arXiv:2308.02583]

[1] Kaiyuan Ji and Eric Chitambar, "Incompatibility as a resource for programmable quantum instruments", *PRX Quantum* **5**, 010340, March 2024. [arXiv:2112.03717]

Conference Proceedings

[2] Kaiyuan Ji, Hemant K. Mishra, Milán Mosonyi, and Mark M. Wilde, "Converse bounds for quantum hypothesis exclusion: A divergence-radius approach", *2025 IEEE International Symposium on Information Theory*, pp. 1–6, June 2025. [arXiv:2501.09712]

[1] Kaiyuan Ji and Eric Chitambar, "Entropic and operational characterizations of dynamic quantum resources", *2023 IEEE International Symposium on Information Theory*, pp. 755–760, June 2023.

Experience

Research Internships

Massachusetts Institute of Technology — Cambridge, Massachusetts, USA
Research Associate, supervised by Seth Lloyd — 07/2025 – 08/2025

University of Illinois Urbana-Champaign — Urbana, Illinois, USA
Visiting Research Assistant, supervised by Eric Chitambar — 06/2021 – 08/2021

Research Visits

Massachusetts Institute of Technology — Cambridge, Massachusetts, USA
Hosted by Seth Lloyd — 13/04/2026 – 23/04/2026

Presentations

Talks

[7] "Fundamental limits for thermodynamic control with quantum feedback", contributed talk at *Beyond IID in Information Theory 14*, Shenzhen, China, June 2026, presented by H. Murray.

[6] "Retrocausal capacity of a quantum channel", contributed talk at *Quantum Resources 2026*, Tokyo, Japan, March 2026, presented by S. Lloyd.

[5] "Converse bounds for quantum hypothesis exclusion: A divergence-radius approach", contributed talk at *ISIT 2025*, Ann Arbor, Michigan, June 2025.

[4] "Fundamental work costs of preparation and erasure in the presence of quantum side information", invited talk at *University of Ulm*, Ulm, Germany, May 2025.

[3] "Postselected communication over quantum channels", contributed talk at *Beyond IID in Information Theory 12*, Urbana, Illinois, July 2024.

[2] "Postselected communication over quantum channels", seminar talk at *Cornell Information Theory Day*, Ithaca, New York, December 2023.

[1] "Entropic and operational characterizations of dynamic quantum resources", contributed talk at *ISIT 2023*, Taipei, Taiwan, June 2023.

Posters

[3] "Fundamental limits for thermodynamic control with quantum feedback", poster at *Quantum Thermodynamics and Decoherence*, Rochester, New York, June 2026.

[2] "Retrocausal capacity of a quantum channel", poster at *Cornell Quantum Day*, Ithaca, New York, November 2025.

[1] "Incompatibility as a resource for programmable quantum instruments", poster at *QIP 2022*, Pasadena, California, March 2022.

Teaching

Cornell University — Ithaca, New York, USA

Foundations of Machine Learning [ECE 3200/5200] — Spring 2026
Teaching assistant, supervised by Ziv Goldfeld

Signals and Systems [ECE 3250] — Fall 2025
Teaching assistant, supervised by David F. Delchamps

Quantum Information Theory (1 lecture) [ECE 6960] — Spring 2025
Interim lecturer (1 lecture), supervised by Mark M. Wilde

Data Science for Engineers [ECE 2720] — Fall 2023
Teaching assistant, supervised by Jayadev Acharya

Fundamentals of Machine Learning [ECE 4200/5420] — Spring 2023
Teaching assistant, supervised by Jayadev Acharya

Press Coverage

"'Back to the future' messages are more efficient", *Physics World*, July 2026.

"Can black holes send information back in time?", *Scientific American*, June 2026.

"We have figured out a new way to send messages into the past", *New Scientist*, April 2026.

Service

Journal Reviewer

New Journal of Physics

Conference Reviewer

2026 IEEE International Symposium on Information Theory (ISIT 2026)
2026 IEEE International Conference on Communications (ICC 2026)
2024 IEEE International Symposium on Information Theory (ISIT 2024)
27th Annual Conference on Quantum Information Processing (QIP 2024)

Updated July 2026